MIT 6.7220/15.084 Nonlinear Optimization (Spring 2026)

Problem Set 4

Due: April 17, 2026 11:59PM ET

1 How are Lagrangian dual of QCQP and SDP relaxation related? (30 points)

Consider the QCQP:

$$p^* = \min_{x \in \mathbb{R}^n} \; x^\top A_0 x + 2b_0^\top x + c_0 \quad \text{s.t.} \quad x^\top A_i x + 2b_i^\top x + c_i \le 0, \quad i = 1, \ldots, m. \tag{1}$$

which can be written as

$$p^* = \min_{x \in \mathbb{R}^n} \bar{x}^\top \bar{A}_0 \bar{x} \quad \text{s.t.} \; \bar{x}^\top \bar{A}_i \bar{x} \le 0, i = 1, \ldots, m, \tag{2}$$

where $\bar{x} = (x^\top, 1)^\top \in \mathbb{R}^{n+1}$ and define $\bar{A}_i \in \mathbb{S}^{n+1}$.

1. (2 points) Prove that the Lagrangian dual function $g(\lambda)$ of the QCQP equals $\inf_{\bar{X} \in \mathcal{R}} \langle \bar{A}_0 + \sum_i \lambda_i \bar{A}_i, \bar{X} \rangle$, where Let $\mathcal{R} = \{\bar{X} \in \mathbb{S}^{n+1} : \bar{X} = \bar{x}\bar{x}^\top, \; \bar{x}_{n+1} = 1\}$.

2. (5 points) Prove that for any linear function $\ell : V \to \mathbb{R}$ in a real vector space V and any nonempty set $S \subseteq V$, $\inf_{x \in S} \ell(x) = \inf_{x \in \mathrm{conv}(S)} \ell(x)$.

3. (15 points) Prove that $\mathrm{conv}(\mathcal{R}) = \{\bar{X} \succeq 0 : \bar{X}_{n+1,n+1} = 1\}$.

   Hint: One direction of set containment is easy. For the harder direction, write $\bar{X} = \begin{pmatrix} A & b \\ b^\top & 1 \end{pmatrix} \succeq 0$ and consider the Schur complement $C = A - bb^\top \succeq 0$ and its eigendecomposition.

4. (3 points) Using (a)–(c), derive the Lagrangian dual of the QCQP, $d_L = \max_{\lambda \ge 0} g(\lambda)$. Show that this coincides with the Lagrangian dual of the SDP relaxation

   $$p^*_{\mathrm{SDP}} = \min_{\bar{X}} \langle \bar{A}_0, \bar{X} \rangle \quad \text{s.t.} \quad \langle \bar{A}_i, \bar{X} \rangle \le 0, \;\; i = 1, \ldots, m, \quad \bar{X}_{n+1,n+1} = 1, \quad \bar{X} \succeq 0,$$

   where only the linear inequalities $\langle \bar{A}_i, \bar{X} \rangle \le 0$ are dualized while $\bar{X} \succeq 0$ and $\bar{X}_{n+1,n+1} = 1$ are kept as set constraints.

5. (5 points) Suppose Slater's condition holds for the SDP relaxation: there exists $\bar{X}_0 \succ 0$ with $(\bar{X}_0)_{n+1,n+1} = 1$ and $\langle \bar{A}_i, \bar{X}_0 \rangle < 0$ for all i. Define the value function

   $$\phi(u) = \inf \; \big\{ \langle \bar{A}_0, \bar{X} \rangle : \langle \bar{A}_i, \bar{X} \rangle \le u_i, \;\; i = 1, \ldots, m, \quad \bar{X}_{n+1,n+1} = 1, \quad \bar{X} \succeq 0 \big\}.$$

   Prove that Slater's condition implies $\phi^{**}(0) = \phi(0)$, thus $d_L = p^*_{\mathrm{SDP}}$. *Hint:* Consult Lecture 10.

   Conclude the *primal characterization*: the Lagrangian dual of the QCQP equals the SDP relaxation value, i.e., $d_L = p^*_{\mathrm{SDP}}$, under Slater's condition. Interpret this: the SDP relaxation is the *primal representation* of the Lagrangian dual of QCQP.

2 Computational: Nonconvex QCQP and SDP relaxation (30 points)

Consider the nonconvex QCQP in $\mathbb{R}^2$:

$$p^* = \min_{x \in \mathbb{R}^2} -x_1^2 - x_2^2 + 4x_1x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

1. (3 points) Plot the contours of the objective function over the feasible region.
2. (2 points) Find all global optimal solution(s) using the contour plot.
3. (10 points) Now we want to prove that the global optimal solution(s) are indeed what you find from visual inspection in the previous question.
   (a) On the boundary circular arc $x_1^2 + x_2^2 = 4, x_1 \geq 0, x_2 \geq 0$, parameterize $x_1 = 2\cos(\theta)$ and $x_2 = 2\sin(\theta)$ for $\theta \in [0, \pi/2]$. Write down the objective function f in terms of θ and simplify it to a function of 2θ using some trigonometric identity such as $2\sin(\theta)\cos(\theta) = \sin(2\theta)$. Find the minimum of f on the circular arc.
   (b) Find the minima of f on the boundary edges $x_1 = 0$ and $x_2 = 0$.
   (c) Find all minimum(s) in the interior of the feasible region, if there exists any. Explain.
   (d) Conclude the global minimum of f.
4. (10 points) Solve the SDP relaxation:
   (a) write down the data matrices $\bar{A}_0$ for the objective, and $\bar{A}_i$ for all the linear constraints and formulate the SDP relaxation with 3-by-3 matrices.
   (b) Solve the SDP relaxation numerically. You may use any language or algorithm you are familiar with.
   We suggest CVXPY with Python. CVXPY is an open source Python-embedded modeling language for convex optimization problems. You can find documentation: `https://www.cvxpy.org`; SDP tutorial: `https://www.cvxpy.org/examples/basic/sdp.html`. Key syntax: `cp.Variable((n,n), symmetric=True)` for a symmetric matrix variable; `X ≫ 0` for $\bar{X} \succeq 0$; `cp.trace(A @ X)` for $\langle A, \bar{X} \rangle$; solvers: SCS or MOSEK.
   Please submit your code and output in a pdf file onto gradescope. In particular, please report the optimal p^*_{SDP}, the optimal solution $\bar{X}^*$, and its rank.
5. (5 points) Formulate and solve the Lagrangian dual of the QCQP as an SDP. Please submit your code. Report the optimal d^* and the dual optimal solution. Does d^* equal to p^*_{SDP}?

3 Minimum Volume Enclosing Ellipsoid (20 points)

Given a finite set of points $x_1, \ldots, x_m \in \mathbb{R}^n$, we are interested in finding an ellipsoid with minimum volume that encloses $x_1, \ldots, x_m$. For an ellipsoid $E = \{x \in \mathbb{R}^n : (x-c)^\top A(x-c) \leq 1\}$ with a positive definite matrix A, the volume of E is given by $|E| = -\det(A)$. The problem can be formulated as

$$\min_{A \in \mathbb{R}^{n\times n}, c \in \mathbb{R}^n} \quad -\log\det(A) \tag{3}$$

$$\text{s.t.} \quad (x_i - c)^\top A(x_i - c) \leq 1, \quad \forall i = 1, 2, \cdots, m, \tag{4}$$

$$A \succeq 0. \tag{5}$$

1. (10 points) Consider an alternative formulation:

$$\min_{M \in \mathbb{R}^{n\times n}, z \in \mathbb{R}^n} \quad -2\log\det(M) \tag{6}$$

$$\text{s.t.} \quad \|Mx_i - z\|^2 \leq 1, \quad \forall i = 1, 2, \ldots, m, \tag{7}$$

$$M \succeq 0. \tag{8}$$

   Prove that if $(\hat{M}, \hat{z})$ is an optimal solution of (7), then $(\hat{A}, \hat{c}) = (\hat{M}^2, \hat{M}^{-1}\hat{z})$ is the optimal solution of (4).

2. (10 points) Solve the problem numerically on an example with $m = 20$ and $n = 2$. You may use any language or algorithm you are familiar with. We suggest to use CVXPY with Python. Please submit your code and output on gradescope.

4 Gradient Descent for Convex Lipschitz Continuous Functions (20 points)

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a differentiable convex function. Rather than being Lipschitz *differentiable*, we only know that f is Lipschitz *continuous*, i.e., there exists $L > 0$ such that

$$|f(x) - f(y)| \leq L\|x - y\|_2, \quad \forall x, y \in \mathbb{R}^n. \tag{9}$$

Let x^* be any optimal solution of $\min_{x\in\mathbb{R}^n} f(x)$ and let $f^* = f(x^*)$. Consider the gradient descent method:

$$x_{t+1} = x_t - h_t \nabla f(x_t). \tag{10}$$

1. (10 points) Prove that for any $T \geq 1$,

$$\sum_{t=0}^{T} h_t(f(x_t) - f^*) \leq \frac{1}{2}\|x_0 - x^*\|^2 + \frac{L^2}{2}\sum_{t=0}^{T} h_t^2. \tag{11}$$

2. (10 points) Suppose we take step size $h_t = \frac{1}{L\sqrt{t+1}}$. Prove that for all $T \geq 1$,

$$\min_{0\leq t\leq T}(f(x_t) - f^*) \leq \frac{L}{2}\frac{\|x_0 - x^*\|^2 + \log(T+1) + 1}{\sqrt{T+1}}. \tag{12}$$

   Note that this convergence rate is much worse than what we have when f is convex Lipschitz *differentiable*. A demonstration of the benefit of stronger smoothness property of the objective function.